

Recommender Systems: Using a Live Data Set for Implementation

This two-part series of articles explains and demonstrates how to implement a recommender system for an online retail store using Python. The first part of the series, published in the May 2022 issue of *Open Source For You*, focused on the theory behind such a system. This second part explains how to implement that system.

The data set used for research can be found and accessed via <https://data.world/uci/online-retail>. Its details are given in the box below.

Solution methodology

So how does one create a system that recommends a specific number of products to users on an e-commerce website, based on their previous purchase history. To construct the answer, we employed the Skip-Gram modelling technique of the word2vec algorithm that we have seen in Part 1. However, we took a customer-focused approach to this.



Source

Dr Daqing Chen, Director: Public Analytics Group (chend '@' lsbu.ac.uk), School of Engineering, London South Bank University, London.

Data set information

This is a transnational data set that contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK based and registered non-store online retail. The company mainly sells unique all-occasion gifts. Many customers of the company are wholesalers.

Attributes used in the data set

InvoiceNo: Invoice number. Nominal -- a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation.

StockCode: Product (item) code. Nominal -- a 5-digit integral number uniquely assigned to each distinct product.

Description: Product (item) name. Nominal.

Quantity: The quantities of each product (item) per transaction. Numeric.

InvoiceDate: Invoice date and time. Numeric -- the day and time when each transaction was generated.

UnitPrice: Unit price. Numeric -- product price per unit in sterling pound.

CustomerID: Customer number. Nominal -- a 5-digit integral number uniquely assigned to each customer.

Country: Country name. Nominal -- the name of the country where each customer resides.